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Digital Twin Framework Proposal

Intern Summer Project

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What is a Digital Twin?

Digital Twin

- Virtual representation of physical object, system, or process
- Uses real-time data to simulate, monitor, and optimize both the counterpart and decisions made in relation to the counterpart
- Data collected and streamed from Internet of Things (IoT) devices

- Useful for...
  - Predictive Modeling
  - Optimizations in processes
  - Streamlining decisions/planning

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What Should the Digital Twin Framework Do?

Goal of the Framework

- The framework should outline a procedure highlighting the incorporation of the Digital Twin for future applied systems and software.
- The framework will be a proposed process for implementation, using a written procedure to implement the digital twin for broad or specific applications.

- Utilization for future applications
- Consider tools for effective operating procedures
- Emulate a ground antenna or satellite
- Utilize Opensource/machine learning AI in design
- Identify individual components/systems for optimization
- Outline process for system monitoring and simulation

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What are the Recommended Use Cases?

Recommended Use Cases

Simulated Combat and Cyber Systems

- Cybersecurity testing and vulnerability assessment
- Electronic warfare simulation
- Mission and battle scenario analysis
- Integration of system analysis
- Future Capabilities

Communication Systems Development

- Network design and optimization
- Network resilience and failure analysis
- Cybersecurity and warfare testing
- Simulated network testing
- Command, control, and communications (C3)

Environmental Effects and Protocols

- Space, weather, and satellite operations
- Network communication and performance under environmental conditions
- Environmental effects on mission readiness

Training Combat Systems

- Mission rehearsal
- Operator, command and control (C2), communication and network, and joint force/multi-domain training
- After-action review and performance analysis

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What are Some Dependencies and Questions?

Needed Resolutions

Digital Twin Framework

Dependencies

<u>Data Dependencies</u>

- Processing, Analytics, integration.

<u>Modeling/Simulation</u>

- Model engines to simulate behavior

<u>Infrastructure</u>

- Cloud platforms

<u>Artificial Intelligence (AI) and Analytic</u>

- Predictive maintenance, failure detection

Dependencies

<u>Sensor Dependencies</u>

- Hardware diagnostics and monitoring

<u>Cybersecurity</u>

- Data transmission, encryption, monitoring

<u>Environmental</u>

- Impact and Prediction

<u>Human Dependencies</u>

- User interface

Questions

- What applications?
- How will AI be integrated?
- Linear or dynamic?
- Safety of implementation?
- How will data be standardized?
- What limitations?
- Any outsourcing?
- What model will be used?

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Associated Risks

Risks

Digital Twin Framework

Scenario 1

If

- Cyber threat occurs

Then

- The high volume of sensitive and interconnected data is vulnerable to leaks, attacks, etc.

Scenario 2

If

- Data is corrupted

Then

- Responses to data sensors and monitoring may be pre-emptive or late

The *Evil* Twin

If

- An unauthorized entity gains control

Then

- They can manipulate the system, which can lead to a decrease in security of the network, falsification of data, and information leaking.

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What Tools, Languages, and Architecture will be Used?

Tools, Languages, Architecture

- Modeling and simulation software tools – Computer Assisted Design (CAD), Unity, etc.
  - Needed for physical and spatial models, as well as simulation
- Programming tools – Linux, Git, C++, etc.
  - Needed for actual development of digital twin
- Data collection and integration tools/methods
  - Necessary for a digital twin synchronize with and simulate a physical counterpart, as well as unify data
- Data storage – Data lakes, data warehouses, time-series databases
  - Needed for data reference and retention
- Artificial Intelligence and/or Machine Learning integration
  - May be useful for optimizing and tracking certain processes (predicting maintenance, data outlier detection, etc.)

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What Deliverables will be Produced?

Deliverables

- A guided framework document outlining the implementation procedure of the digital twin
- A basic prototype system (model) demonstrating how the digital twin works
- A presentation based on the basic prototype and framework document
- A diagram demonstrating the how the digital twin framework integrates different systems
- A data collection system/program demonstrating proposed digital data collected by the digital twin
- What applications should be built first (Recommendations)
- An appendix of software's that are necessary for the creation digital twins
- An analysis of how the framework suits Auria’s work and mission statement

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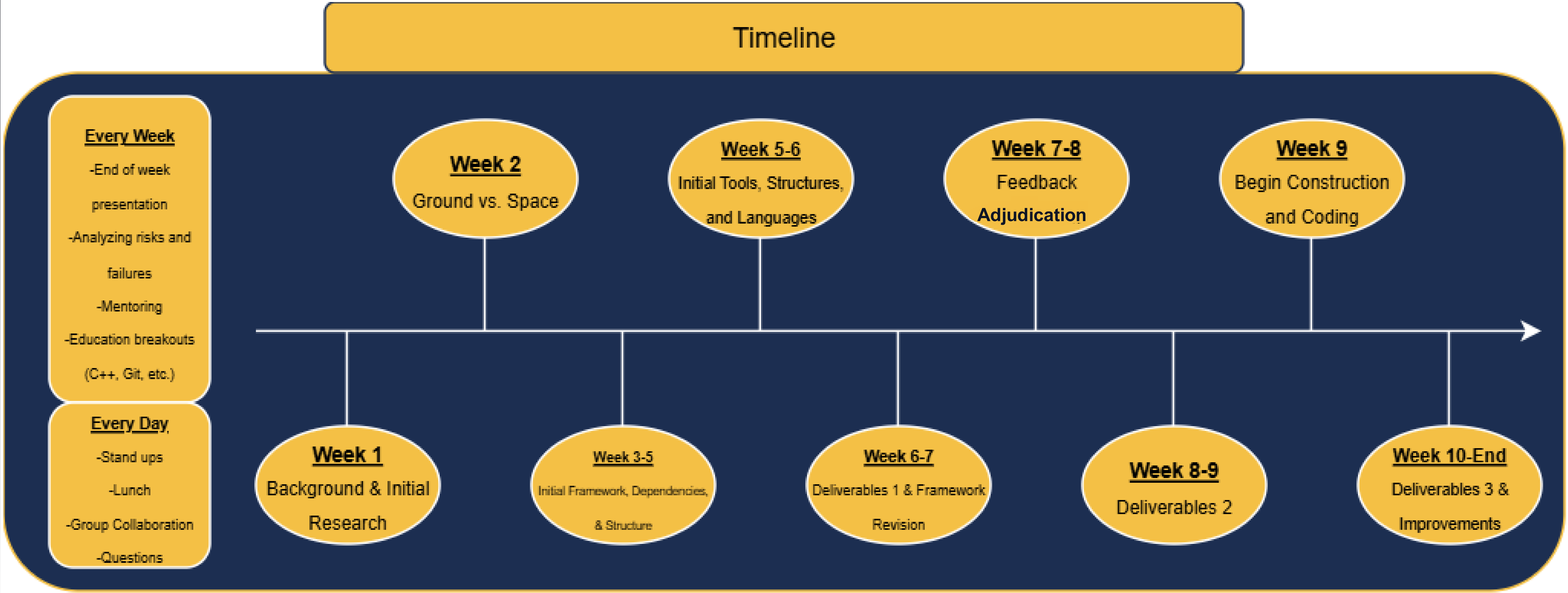
Proposed Timeline

Proposed Timeline



Deliverables 1, 2, 3, … = Prototype Frameworks Ready for Review and Revision

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Auria's Mission

Rising Together

Elevate

- The framework concept *Elevates* the technological integration of AI and Auria's products by creating a preventative and analytical system for efficient data collection.

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- The Digital Twin Framework will move the company forward by developing new adaptive technologies capable of testing all probabilities/scenarios a system may face.

Human-Centered

- This project will engage the Interns with engineering systems and provide valuable insight for company protocol going forward.

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Resources

- Evil digital twins and other risks: the use of twins opens up a host of new security concerns | CSO Online
- Digital Twins For A Digital World: Data-Driven Training Optimizing The Ready Medical Force
- The Digital Twin Paradigm for Future NASA and U.S. Air Force Vehicles - NASA Technical Reports Server (NTRS)
- Non-Technical Intro to Digital Twins | Towards Data Science
- What is digital-twin technology? | McKinsey
- What Is a Digital Twin? | NVIDIA Glossary
- What is Digital Twin Technology? - Digital Twin Technology Explained - AWS
- What is IoT? - Internet of Things Explained - AWS
- Digital twins in manufacturing & product development | McKinsey
- Digital twin: Data exploration, architecture, implementation and future - ScienceDirect
- Digital Twin of Space Environment: Development, Challenges, Applications, and Future Outlook
- [2508.05717] On Digital Twins in Defense: Overview and Applications
- Digital Twin Framework Explained
- Twinning for Space-Air-Ground-Sea Integrated Networks: Beyond Conventional Digital Twin Towards Goal-Oriented Semantic Twin

- Digital Twins For A Digital World: Data-Driven Training Optimizing The Ready Medical Force

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